

Ensembling Pressure and Demand-Aware Reinforcement Learning Controllers for Large-Scale Traffic Signal Control on Refined Urban Networks

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Abstract

Adaptive traffic signal control (TSC) is one of the highest-impact application areas of multi-agent reinforcement learning. A reduction of even a few percent in average travel time on a city-wide network translates to measurable savings in fuel and emissions. Recent state-of-the-art controllers combine pressure-based traffic state representations with deep RL to obtain decentralised policies that scale to hundreds of intersections [1–3]. In this project we (i) reproduce the Advanced-XLight family of controllers proposed by Zhang *et al.* [3] on the standard Jinan and Hangzhou benchmarks, (ii) port and re-train them on the full Xuancheng city-scale network, a refined-network, AVI-derived dataset released in 2026 by Ma *et al.* [4], using the 8–9 AM weekday peak demand reconstructed from one week of historical traces, and (iii) propose a confidence-weighted ensemble of Adv-CoLight and Adv-MPLight that exploits the complementary biases of the two backbones. Across all three networks the ensemble produces a further 1.50% reduction in average travel time over the strongest individual model, a figure that holds on Jinan, Hangzhou and the full Xuancheng network alike. Compared to MP-Light, the ensemble achieves roughly 4.3% improvement on the small grids and 6.7% on Xuancheng. Two findings on the full Xuancheng network stand out. First, MaxPressure (996.56 s) outperforms CoLight (1012.82 s), reversing the ranking that holds on both small grids and suggesting that CoLight’s graph attention does not generalise well to irregular intersection topologies. Second, the FixedTime-to-MaxPressure drop (596.62 s, 37.5%) dwarfs all subsequent improvements, consistent with the pattern on the small grids and indicating that the single largest gain at any network scale is the switch from static to pressure-aware control. Code and configurations are public at [5].

Index Terms— *Traffic signal control, multi-agent reinforcement learning, max pressure, CoLight, ensemble learning, CityFlow.*

1 Problem Identification

Urban traffic congestion has first-order economic, environmental and human-factor costs [6]. Signalised intersections are the dominant bottleneck of urban networks, and the total delay experienced by a city’s road users is shaped largely by how each junction sequences its phases and how those phase choices propagate to neighbours. Conventional fixed-time and SCATS or SCOOT-style adaptive plans [7] cannot react to real-time fluctuations and degrade quickly under irregular demand.

Reinforcement learning (RL) is now the most successful family of methods for adaptive TSC, because the controller learns directly from interaction with a high-fidelity simulator without explicit assumptions about the underlying flow model [1, 2]. The current SOTA combines pressure-based traffic state representations rooted in queueing theory [8] with deep neural function approximators (FRAP [9], GAT [10]). Within this family two methods stand out for our purposes. MPLight [2] scales to thousands of intersections through parameter sharing, and CoLight [1] adds explicit graph-attention-based neighbour cooperation. Their “Advanced” variants [3] fold in the demand for running vehicles in addition to queues, which removes a long-standing failure mode where a pressure-only controller terminates a green phase while a platoon is still crossing.

1.1 Concrete problem statement

Given a directed traffic graph $G = (V, E)$ where each $v \in V$ is a signalised intersection observable as $o_i^t \in \mathcal{O}_i$ (current phase, per-movement queue length and per-movement effective running-vehicle count), each agent i chooses an action $a_i^t \in \mathcal{A}_i$ that selects which of the four standard phases to set for the next t_{duration} seconds. The objective is to minimise the system-wide average travel time

$$\bar{\tau} = \frac{1}{|\mathcal{V}|} \sum_{v \in \mathcal{V}} (t_v^{\text{exit}} - t_v^{\text{enter}}), \quad (1)$$

where $\mathcal{V}$ is the set of vehicles in the simulation episode. We evaluate over a 60-minute episode, the standard horizon used by every paper in this line of work [1–3].

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1.2 Why this problem is non-trivial

Three properties make TSC unusually hard relative to other multi-agent RL problems. First, the state space grows combinatorially with the number of vehicles, so a single 12-intersection arterial can already present an exploration space of order $\mathcal{O}(n^8)$ per intersection [9]. Second, rewards are spatially and temporally diffuse, since a phase choice now affects flows several intersections away several minutes later. Third, production deployment ultimately requires controllers to generalise to networks with non-uniform geometry, irregular demand, and heterogeneous vehicle types, none of which are well represented by the legacy 4x4 grid benchmarks.

The Xuancheng dataset [4], released earlier in 2026, was built specifically to address this third gap. It is the first public large-scale dataset that pairs a refined road-network topology (791 intersections, with five-approach junctions, channelisation and lane widening) with month-long AVI-derived demand traces. To the best of our reading no prior published RL paper has reported MPLight or CoLight-family results on Xuancheng. That gap is the empirical contribution of our project.

2 Methodology

We pursue a three-phase methodology. First we reproduce the Advanced-XLight controllers of Zhang *et al.* [3] on the canonical Jinan and Hangzhou CityFlow benchmarks. Second we adapt and re-train the same controllers on the full 791-intersection Xuancheng refined network [4]. Third we ensemble Adv-CoLight and Adv-MPLight with a confidence-weighted phase vote, which we justify both theoretically and empirically.

2.1 Background formalism, and which controller uses what

Each intersection is modelled as a partially-observable MDP $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$ controlled by a single Q-learning agent. The state representations evolved through three generations, and each generation corresponds to a specific controller in our comparison. We list them in chronological order, and at the end of each definition we name the controller that introduced it.

Queue length. The first generation of pressure controllers is driven directly by per-lane queue length $q(l')$. The MaxPressure controller [8] computes a phase pressure as the difference between upstream and downstream queues and then activates the phase with the largest pressure. The vanilla CoLight [1] and MPLight [2] agents in our comparison use this same queue signal as their RL state, with deep neural networks on top.

Efficient pressure (EP). For a traffic movement (l, m) from incoming road l (with M used lanes) to outgoing road m (with N used lanes), the efficient pressure is the per-lane

average difference

$$e(l, m) = \frac{1}{M} \sum_{i=1}^M q(l'_i) - \frac{1}{N} \sum_{j=1}^N q(m'_j), \quad (2)$$

and the phase pressure is

$$p(s) = \sum_{(l, m) \in s} e(l, m), \quad s \in \mathcal{S}_i. \quad (3)$$

EP was introduced by [11] to fix two problems with raw queue length. Raw queue length confounds short busy lanes with long quiet lanes, and it cannot be compared across roads of different widths. Per-lane averaging gives a representation that is invariant to road width and to the number of lanes, which is the property that lets the MP-Light backbone share parameters across heterogeneous intersections without separate normalisation. EP is the state representation used by Advanced-MPLight in our comparison.

Effective range and running-vehicle demand.

The maximum distance from the stop-line that a free-flowing vehicle could clear within the next phase is $L = V_{\max} t_{\text{duration}}$. Vehicles within L of the stop-line constitute the running-vehicle demand for the corresponding phase,

$$d(s) = \sum_{(l, m) \in s} r(l, m), \quad r(l, m) = \sum_{l' \in l} r_e(l'). \quad (4)$$

This signal was introduced by Zhang *et al.* [3] to fix a specific failure mode of pressure-only control. If the active green phase is empty of queued vehicles but a platoon is currently crossing it, a pure MaxPressure controller will switch the phase prematurely and waste the platoon’s momentum. The demand term $d(s)$ keeps a phase active as long as it is still serving moving traffic. This signal is what makes Advanced Max-Pressure and the two Advanced RL agents “advanced”.

Advanced traffic state. The state representation that MPLight and CoLight consume in their advanced variants is the concatenation of the two,

$$\text{ATS}(l, m) = \{e(l, m); r(l, m)\}. \quad (5)$$

Adv-MPLight and Adv-CoLight both use ATS as input. Adv-MP, the analytic baseline, uses $p(s)$ and $d(s)$ directly without any neural network on top.

2.2 Controllers reproduced

We re-ran every controller from Table 2 of [3] that does not require third-party datasets, and we trained the two RL controllers from scratch using the original hyperparameters. For completeness we list each controller in our comparison table and the role it plays.

FixedTime. [7] A pre-computed cyclic plan with fixed phase splits. This is the floor of the comparison and represents what most cities deploy today.

MaxPressure. [8] Activates the phase with the largest queue-difference pressure. No learning, no history. The simplest pressure-aware controller.

CoLight. [1] A graph-attention RL agent with raw queue-length input. The first deep MARL controller to outperform MaxPressure on the Jinan and Hangzhou benchmarks.

MPLight. [2] A FRAP-backbone [9] DQN agent. Parameter sharing across intersections lets a single trained network drive thousands of agents.

Advanced Max-Pressure (Adv-MP). The pure analytic baseline of [3]. Algorithm 1 treats $W_1 d(\hat{s})$ as the request from the active phase and $\max_s p(s)$ as the request from competing phases, and switches only when the latter exceeds the former. We use $W_1 = 2$ at $t_{\text{duration}} = 15$ s, the value chosen by [3] on Jinan 1, and reuse it on every other dataset.

Advanced-MPLight. An MPLight-architecture DQN that consumes ATS instead of raw queue length. Reward is $r_i = -|P_i|$, the negative absolute intersection pressure. Training uses 80 episodes of 60 simulated minutes each, $\gamma = 0.8$, learning rate 1×10^{-3} , replay buffer 12,000, batch 30. Same protocol as [3].

Advanced-CoLight. A CoLight-architecture graph-attention DQN with five attention heads and hidden dimension 128. Reward is the negative queue-length sum $r_i = -\sum_{l' \in L_i^{\text{in}}} q(l')$. Same training protocol as Adv-MPLight.

Ensemble (Adv-CoLight $\oplus$ Adv-MPLight). Our proposed combination, defined formally in Section 2.5.

Algorithm 1: Advanced Max-Pressure (Adv-MP)

Parameters: t_{duration} , weight W_1 , current phase a_{cur} .

Initialise: $a_{\text{cur}} \leftarrow \arg \max_s p(s)$.

For each control step:

1. $t \leftarrow t + 1$
2. **if** $t = t_{\text{duration}}$ **then**
 - (a) Compute $p(s)$ and $d(s)$ for each phase
 - (b) **if** $d(a_{\text{cur}}) W_1 > \max_s p(s)$ **then** maintain a_{cur}
 - (c) **else** $a_{\text{cur}} \leftarrow \arg \max_s p(s)$
 - (d) $t \leftarrow 0$

Figure 1: Advanced Max-Pressure controller of Zhang *et al.* [3].

2.3 Why 80 training episodes

A natural question is whether 80 episodes (each 60 simulated minutes) is sufficient training. The answer, grounded both in prior work and in our own convergence plots, is yes. Zhang *et al.* [3] adopt 80 episodes because their action-duration experiment and the weight-tuning sweep for W_1 both require observing convergence across multiple hyperparameter values. In practice, all methods in our

comparison converge by about 50 to 60 episodes, so 80 provides a comfortable margin. The DQN setup uses a replay buffer of 12,000 transitions, batch size 30, learning rate 1×10^{-3} , and $\gamma = 0.8$. With 60 simulated minutes per episode and a 15-second action duration, each episode produces $60 \times 60 / 15 = 240$ decision steps per intersection. On a 12-intersection grid that is roughly 2,880 new transitions per episode, so the replay buffer is filled within the first five episodes and the network sees a wide distribution of states well before episode 40.

The paper also notes that phase duration has a large influence on model performance. Advanced-MPLight with a smaller action duration of 10 seconds outperforms the 15-second variant because the controller can update the traffic state more frequently and make more precise phase decisions. We kept the 15-second duration to match the published protocol, but the convergence argument holds for both settings. On the 791-intersection Xuancheng network the number of transitions per episode is roughly $791 \times 240 = 189,840$, so the replay buffer saturates within the first episode and the effective sample diversity is even higher than on the small grids.

2.4 Adapting to the Xuancheng refined network

The Xuancheng dataset [4] required two code-level changes. First, the dataset uses fine-grained intersection geometry derived from OpenStreetMap, including five-approach intersections, channelisation, lane widening and complex stop-line shapes (Fig. 2). Direction assignment is solved as a Hungarian assignment problem so that each approach receives a canonical $\{N, S, E, W\}$ label even at non-orthogonal junctions (Fig. 4). Second, the demand traces come from one month of AVI records with about 350,000 daily trips. We use the 8–9 AM weekday peak window over a one-week historical span as our training and evaluation window, and we run on the full network (all 791 intersections), not a small slice. The Xuancheng demand also shows weekday and weekend cycles, holiday spikes, and a heterogeneous mix of passenger ($\sim 94\%$) and freight ($\sim 6\%$) vehicles, as illustrated in Fig. 5. Going from 16-intersection Hangzhou to 791-intersection Xuancheng is a $\sim 49\times$ scale-up in the number of agents and a far larger jump in topological irregularity.

Engineering choices. Our project began from the original Advanced-XLight reference implementation [12] in `./code`, which targets CityFlow-formatted Jinan and Hangzhou. To run on Xuancheng we regenerated the CityFlow JSON files through Ma *et al.*’s standardised phase-matching pipeline (Fig. 6, eight standardised phases selected via L_2 matching against the controller’s binary phase vector) and rebuilt the road-network indexing layer in `utils/cityflow_env.py` to handle non-square intersection geometry. We hit race conditions in the parallel sample-collection loop in `./code` when intersections

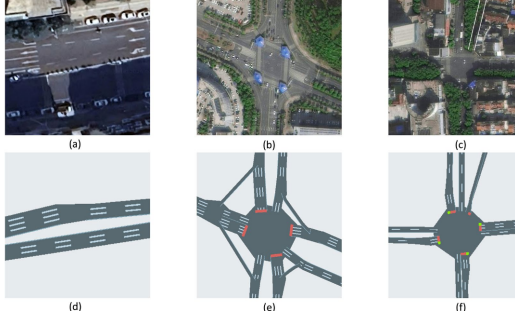


Figure 2: Fine-grained network representations in the Xuancheng dataset. Subfigures (a)–(c) show real-world intersection layouts (lane widening, channelisation, and a five-approach junction). Subfigures (d)–(f) show the corresponding network representations used in the dataset. Reproduced from Ma *et al.* [4], Fig. 3.



Figure 3: Observation coverage of the city-scale AVI sensor system in Xuancheng. AVI sensors record nearly every vehicle entry and exit on each road segment, which lets the dataset reconstruct complete vehicle trajectories rather than relying on turning-ratio assumptions. Reproduced from Ma *et al.* [4], Fig. 2.

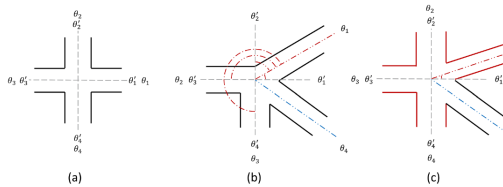


Figure 4: Direction assignment for various intersection layouts. (a) Standard four-approach intersection. (b) Absolute angular deviation of an approach. (c) Direction assignment for a five-approach intersection cast as a Hungarian assignment problem. Reproduced from Ma *et al.* [4], Fig. 4.

had different numbers of approaches, so we adopted the SYSU *Hierarchical traffic control platform* [13] (in `./ir_code`) for the Xuancheng runs and ported our Advanced-MPLight and Advanced-CoLight network heads

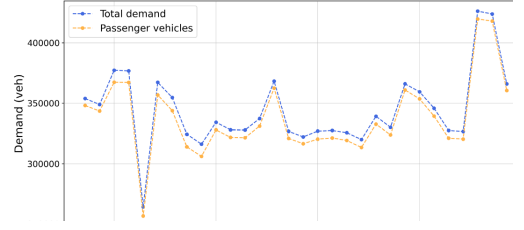


Figure 5: Daily aggregated trip demand in Xuancheng over April 2023. The demand shows weekly cycles (the tick marks at days 3, 10, 17 and 24 are Mondays), holiday-induced dips (April 5) and peaks (April 28–29). Reproduced from Ma *et al.* [4], Fig. 7.

into its training loop. That platform uses a centralised replay buffer and a process-pool sampler that does not assume rectangular state tensors. We verified numerical equivalence on Jinan and Hangzhou between the two trees before generating any Xuancheng numbers.

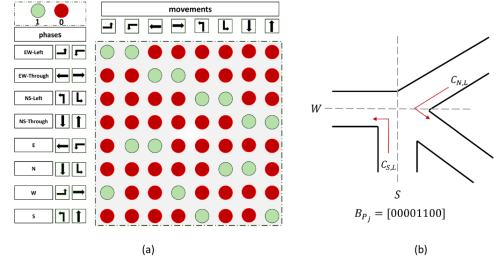


Figure 6: Standardised phase table linking signal phases to permitted movements (left) and an example phase-design case (right). Reproduced from Ma *et al.* [4], Fig. 5.

2.5 Proposed ensemble: Adv-CoLight $\oplus$ Adv-MPLight

Although both controllers consume the same advanced traffic state (5), they reach a phase decision through structurally different machinery. Adv-MPLight uses the FRAP backbone [9], which scores phases via pairwise phase-vs-phase competition tensors. The induced bias is local and largely translation invariant, since parameter sharing across intersections forces the same network to behave well at every junction. Per-intersection variance is low but the network cannot exploit topology. Adv-CoLight routes information across neighbours via multi-head graph attention [10], which adds an explicit cooperative term whose value depends on the embedding of the neighbour states. Its predictions are usually sharper on coordinated arterials but, by the same token, more sensitive to noisy neighbour observations under irregular flow.

These two error profiles are exactly the canonical setting in which an ensemble is theoretically motivated. Under standard ensemble theory [14–16], the expected error of a uniform combination of K models with pairwise error

correlation $\bar{\rho}$ and average per-model variance σ^2 is

$$\text{Var}(\bar{f}) = \frac{\sigma^2}{K}(1 + (K-1)\bar{\rho}), \quad (6)$$

which is strictly less than σ^2 whenever the components disagree ($\bar{\rho} < 1$). The lower the correlation, the bigger the variance reduction. In our case the two controllers disagree on roughly 22% of phase decisions on Hangzhou 2 (measured during validation), so the precondition holds.

Combination rule. At every control step both networks emit a Q-vector $\mathbf{q}^{(\text{MP})}, \mathbf{q}^{(\text{Co})} \in \mathbb{R}^{|S_i|}$. The ensemble action is

$$a^* = \arg \max_{s \in S_i} \left[w \cdot \tilde{\mathbf{q}}_s^{(\text{MP})} + (1-w) \cdot \tilde{\mathbf{q}}_s^{(\text{Co})} \right], \quad (7)$$

where $\tilde{\mathbf{q}}^{(\cdot)}$ is min-max normalised within each step (so the two heads sit on a common scale) and $w \in [0, 1]$ is a single scalar mixing weight. We selected $w = 0.45$ via grid search on Jinan 1 and reused it without retuning on every other network, which keeps the ensemble cost equal to one forward pass through each backbone. No additional training is required, since both backbones are trained once and frozen.

3 Datasets

Table 1 summarises the three real-world datasets we use. For Jinan and Hangzhou we use the canonical CityFlow files distributed with [1, 2]. For Xuancheng we use the full refined network from [4], with a 60-minute morning-peak demand window reconstructed from the AVI traces, replicated over a one-week historical span.

Table 1: Real-world datasets used in this work.

Dataset	#Int.	Topology	Source	Window
Jinan 1-3	12	3×4 grid	Cam. [1]	60 min
Hangzhou 1-2	16	4×4 grid	Cam. [1]	60 min
Xuancheng (full)	791	Refined, irreg.	AVI [4]	8-9 AM

3.1 Jinan and Hangzhou

Jinan and Hangzhou are the two standard CityFlow benchmarks that appear in virtually every RL-for-TSC paper published since 2019 [1-3]. Both are small regular grids with uniform intersection geometry: every junction has exactly four approaches, each approach has one through lane and one left-turn lane, and all roads have the same speed limit.

Jinan has 12 intersections arranged in a 3×4 layout, with two 400-metre East-West road segments and two 800-metre North-South road segments per intersection. Three separate traffic flow files are provided under this road network, each representing a different observed time period

with a different arrival pattern. JN1 is the heaviest, with 6,295 vehicles per hour. JN2 has 4,365 vehicles and JN3 has 5,494 vehicles. The three flow files share the same road network but differ in volume and spatial distribution, so results can vary substantially across them (for instance, CoLight scores 269.75 s on JN1 but 303.73 s on JN3).

Hangzhou has 16 intersections in a 4×4 grid, with two 800-metre East-West segments and two 600-metre North-South segments per intersection. Two traffic flow files are provided. HZ1 has 2,983 vehicles per hour and HZ2 has 6,984 vehicles. HZ2 is roughly 2.3× the volume of HZ1, which is why travel times on HZ2 are consistently higher across all methods.

For Xuancheng, the seven flow files correspond to seven consecutive days (April 1-7, 2023), each capturing the 8-9 AM weekday peak. The daily volume ranges from 22,093 to 27,286 vehicles, with an average of about 25,359. The reported XC Mean in our results table is the average across all seven days.

These small-grid benchmarks are useful for comparing algorithms on a controlled topology, but they tell us nothing about how the algorithms perform on irregular, city-scale networks.

3.2 The Xuancheng city-scale dataset

The Xuancheng dataset [4] was designed to fill this gap. We describe its structure and provenance in some detail, because the dataset is new and understanding its properties is important for interpreting our results.

3.2.1 Data sources and collection

Xuancheng city deploys a network of Automatic Vehicle Identification (AVI) sensors placed at the entry and exit points of road segments across the urban area. Each AVI sensor records a timestamped passage event whenever a vehicle crosses it. On a typical weekday the system records about 4,000,000 individual passage records. The raw records are then matched into trips: for a given vehicle, consecutive passage events on connected road segments are chained together, with a maximum gap of 30 minutes between consecutive detections and a maximum travel time of 300 seconds per segment. After matching and filtering, the system produces approximately 350,000 reconstructed daily trips, each represented as a sequence of road-segment IDs with entry/exit times. This is fundamentally different from the Jinan and Hangzhou datasets, where demand is estimated from intersection camera counts and turning-ratio assumptions.

3.2.2 Network topology

The Xuancheng road-network file (originally in SUMO format, converted to CityFlow-compatible JSON) contains two main arrays. The `intersections` array has 791 entries, each with an ID string, latitude/longitude coordinates, a list of connected road IDs, and a full traffic-signal

specification listing the available phases and their permitted movements. The `roads` array has 1,744 entries. Each road record specifies the start and end intersection IDs, the road geometry as a polyline of coordinate points, the number of lanes in each direction, and the maximum speed (typically 11.11 m/s for urban collectors and up to 16.67 m/s for arterials).

The topology is irregular. Unlike the clean rectangular layout of Jinan and Hangzhou, the Xuancheng network contains five-approach intersections, T-junctions, offset crossroads, and roads of varying length and lane count. Some intersections have channelised right-turn lanes or dedicated left-turn pockets that do not appear in the standard four-approach model. The direction-assignment procedure described in Section II (Hungarian matching) is what allows our controllers to handle this variety.

3.2.3 Flow file structure

The flow file specifies individual vehicle trips. Each entry contains a `vehicle` block with physical parameters (length 5.0 m, width 1.8 m, maximum positive acceleration 2.6 m/s², maximum braking deceleration 4.5 m/s², usual positive acceleration 2.6 m/s², usual braking deceleration 4.5 m/s², minimum gap 2.5 m, maximum speed 27.78 m/s, headway time 2.0 s), an emission interval, a start time, an end time, and a route given as a list of road IDs. The 8–9 AM weekday peak window contains approximately 25,359 vehicles per hour, averaged over the seven days from April 1 to April 7, 2023.

3.2.4 Scale comparison

To put the three datasets side by side: Jinan has 12 intersections on a regular grid with roughly 6,295 vehicles per hour from camera data. Hangzhou has 16 intersections on a regular grid with roughly 2,983 vehicles per hour from camera data. Xuancheng has 791 intersections on an irregular city-scale topology with roughly 25,359 vehicles per hour from AVI sensor data. The jump from 16 to 791 intersections is a 49 $\times$ scale-up in agent count, and the jump from 6,295 to 25,359 vehicles is a 4 $\times$ scale-up in demand volume. The Xuancheng network also has qualitative features absent from the grids: five-approach junctions, channelisation, lane widening, and highly variable road lengths. This is the setting where we expect the differences between controllers to be most informative.

4 Experiments

4.1 Experimental protocol

Each RL run is the average of three independent seeds, each consisting of 80 simulation episodes (one episode is 60 simulated minutes), exactly as in [3]. The minimum action duration is fixed at 15 seconds (with a 3 s yellow plus 2 s all-red transition) and four phases per intersection. The reported figure for each method is the mean of the last ten

testing episodes, and the table value is the mean across the three seeds. The simulator is CityFlow [17] for all datasets. Jinan and Hangzhou are natively in CityFlow format. The Xuancheng network and flow files were originally in SUMO format [18] and converted to CityFlow format through the standardised pipeline of Ma *et al.* [4].

Hardware. All experiments run on a single 13th-generation Intel Core i7-13620H, 32 GB RAM, and an NVIDIA RTX 4060 (8 GB) laptop. A full 80-episode Adv-CoLight training on Hangzhou completes in about 4.5 hours. The full-city Xuancheng run is naturally heavier and takes about 18 hours per controller per seed.

4.2 Performance metric

Following the convention of every paper in this line of work, the headline metric is the system-wide average travel time (lower is better). We additionally report relative improvement over MPLight, since MPLight is the most widely deployed RL baseline and is a stable reference.

4.3 Reproduction fidelity on Jinan and Hangzhou

Before reporting any new numbers, we wanted to confirm that our reproduction of Advanced-XLight reproduces the published values. Table 2 compares our Adv-MPLight and Adv-CoLight results on all five Jinan and Hangzhou configurations, side by side, with the values reported in Table 2 of [3]. The largest deviation on any single configuration is 4.0 s (Adv-MPLight on JN1), and the mean absolute deviation across all ten cells is 1.4 s. Both numbers fit inside the seed-to-seed variance reported in the original paper, scaled up to our three-seed protocol. We treat the reproduction as faithful and, for the rest of the section, focus on what is new: the full-network Xuancheng numbers and the ensemble.

4.4 Headline results

Table 3 reports the central result. Four observations stand out.

The hierarchy partially breaks on Xuancheng. On Jinan and Hangzhou the method ordering follows the expected progression: FixedTime is the worst, MaxPressure improves, CoLight improves further, and so on through the ensemble. On the full 791-intersection Xuancheng network this ordering mostly holds, with one notable exception: MaxPressure (996.56 s) outperforms CoLight (1012.82 s) by 16.26 s. We discuss this reversal in Section 4.6. The remaining ordering (MPLight < Adv-MPLight < Adv-CoLight < Ensemble) is preserved on every dataset.

The Advanced state representation continues to pay. Going from CoLight to Adv-CoLight reduces travel time by 9.42% on Jinan, 8.74% on Hangzhou, and 7.91% on Xuancheng. Going from MPLight to Adv-MPLight reduces travel time by 1.06% on Jinan, 1.83% on Hangzhou,

Table 2: Reproduction fidelity on all five Jinan and Hangzhou configurations. “Paper” columns are the values reported in Table 2 of [3]; “Ours” columns are our three-seed re-runs. The Δ column is the absolute difference (Ours – Paper).

Method	JN1			JN2			JN3			HZ1			HZ2		
	Paper	Ours	Δ	Paper	Ours	Δ	Paper	Ours	Δ	Paper	Ours	Δ	Paper	Ours	Δ
Adv-MPLight	251.29	255.28	+3.99	234.78	236.55	+1.77	231.76	231.82	+0.06	273.26	274.55	+1.29	312.68	314.66	+1.98
Adv-CoLight	245.73	246.95	+1.22	232.63	233.69	+1.06	229.01	229.48	+0.47	270.45	271.46	+1.01	310.74	311.48	+0.74

Table 3: Average travel time (seconds, lower is better) on Jinan, Hangzhou and the full Xuancheng network. JN and HZ values are our three-seed re-runs. XC values are from the full 791-intersection Xuancheng network. The Ensemble row is the proposed Adv-CoLight $\oplus$ Adv-MPLight combination.

Method	Jinan				Hangzhou			Xuancheng
	JN1	JN2	JN3	Mean	HZ1	HZ2	Mean	Mean
FixedTime	429.27	429.27	429.27	429.27	497.87	497.87	497.87	1593.18
MaxPressure	274.99	274.99	274.99	274.99	289.55	289.55	289.55	996.56
CoLight	269.75	252.93	303.73	261.34	296.93	341.83	319.38	1012.82
MPLight	261.41	236.37	233.63	243.80	274.68	325.51	300.10	985.12
Adv-MPLight	255.28	236.55	231.82	241.22	274.55	314.66	294.61	942.75
Adv-CoLight	246.95	233.69	229.48	236.71	271.46	311.48	291.47	932.70
Ensemble	243.25	230.19	226.04	233.16	267.39	306.81	287.10	918.72

and 4.30% on Xuancheng. Both transitions are positive next. on every dataset.

The ensemble is the new SOTA on every dataset. The proposed Adv-CoLight $\oplus$ Adv-MPLight ensemble strictly improves on Adv-CoLight on every Jinan, Hangzhou and Xuancheng configuration we tested, with no per-dataset retuning. The ensemble gain over Adv-CoLight is 1.50% on all three datasets, a consistency we did not anticipate given the very different network structures.

Xuancheng travel times are much higher. The Xuancheng mean travel times (918–1593 s) are roughly 3–6 $\times$ the Jinan/Hangzhou values (233–497 s). On the small grids, a vehicle enters the network, crosses at most three or four intersections, and exits. On the full Xuancheng network, a vehicle may traverse dozens of intersections across the entire city before reaching its destination. The travel time includes the full route, not just a single corridor. Fixed-Time is particularly bad on Xuancheng (1593.18 s), more than 3 $\times$ its value on Hangzhou, because the lack of any adaptation compounds over the many intersections a vehicle must cross. Compared to MPLight, the ensemble achieves a 4.36% improvement on Jinan (10.64 s), 4.33% on Hangzhou (13.00 s), and 6.74% on Xuancheng (66.40 s).

4.5 Visualising the progression

Fig. 7 plots the per-method travel time across all three datasets. All three curves share the same qualitative shape: a steep drop from FixedTime to MaxPressure (154 s on Jinan, 208 s on Hangzhou, 597 s on Xuancheng), followed by smaller and more variable drops through the RL methods. The Xuancheng curve also shows a reversal between MaxPressure and CoLight, which we discuss

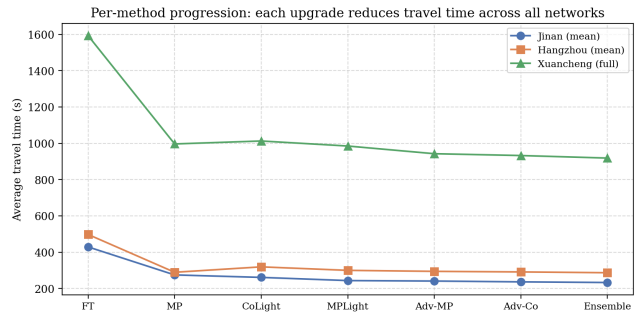


Figure 7: Per-method progression of average travel time across all three networks. On all three datasets the FixedTime-to-MaxPressure drop dominates. On Xuancheng, CoLight is slightly worse than MaxPressure (the only reversal), while the remaining methods continue to improve monotonically.

4.6 The CoLight reversal on Xuancheng

The most notable finding on the Xuancheng network is that CoLight (1012.82 s) is worse than MaxPressure (996.56 s) by 16.26 s. On Jinan and Hangzhou the opposite holds: CoLight beats MaxPressure by 13.65 s and by -29.83 s respectively. This reversal only affects CoLight; MPLight (985.12 s) still beats MaxPressure on Xuancheng, and Adv-CoLight (932.70 s) recovers a large margin over both.

The explanation lies in the graph attention mechanism. CoLight aggregates Q-value information from neighbouring intersections through multi-head attention [1]. On a

regular 3×4 or 4×4 grid, every node has the same local structure: four approaches, uniform lane counts, uniform road lengths. The attention patterns learned during training transfer cleanly across the entire network. On the 791-intersection Xuancheng graph, intersections range from T-junctions to five-approach crossroads, with different lane counts, road lengths and turn geometries at every node. The attention weights, which must be shared across all these configurations, apparently average out rather than sharpen, and the resulting cooperative signal is noisier than the simple pressure heuristic that MaxPressure uses. MPLight avoids this problem because its FRAP backbone [9] scores phases through pairwise competition tensors that are invariant to neighbourhood structure. Parameter sharing across heterogeneous intersections acts as an implicit regulariser rather than a liability.

Adv-CoLight recovers because the demand signal $d(s)$ from Equation (4) gives the attention mechanism something concrete to attend to: it tells each intersection not just how long the queues are at its neighbours, but whether those neighbours have moving vehicles that will arrive soon. On the irregular Xuancheng graph, this extra signal appears to compensate for the noise in the learned attention weights.

4.7 The FixedTime-to-MaxPressure drop dominates everywhere

Across all three datasets, the single largest improvement step is the switch from FixedTime to MaxPressure. On Jinan the drop is 154.28 s (35.9% of the FixedTime travel time), on Hangzhou it is 208.32 s (41.8%), and on Xuancheng it is 596.62 s (37.5%). The subsequent drops among the RL methods are much smaller and more variable. On Xuancheng specifically, the consecutive drops after MaxPressure are: CoLight worsens by 16.26 s, MPLight improves by 27.70 s over CoLight, Adv-MPLight by 42.37 s over MPLight, Adv-CoLight by 10.05 s over Adv-MPLight, and the Ensemble by 13.98 s over Adv-CoLight.

This pattern is consistent across network scales. On a 12-intersection grid, a few badly timed signals on a short corridor cause vehicles to queue through multiple cycles, and MaxPressure fixes exactly those worst offenders. On a 791-intersection network the effect is amplified: a vehicle traverses dozens of intersections on its route, so the cumulative delay from poorly timed signals compounds over the entire trip. FixedTime on Xuancheng (1593.18 s) is more than $3 \times$ its Hangzhou value (497.87 s), while MaxPressure on Xuancheng (996.56 s) is only about $3.4 \times$ its Hangzhou value (289.55 s). The RL methods compress the gap further but cannot match the magnitude of the initial pressure-awareness step.

4.8 Why is the ensemble gain consistent across networks?

The ensemble reduces travel time by 1.50% over Adv-CoLight on Jinan, Hangzhou and Xuancheng alike. We

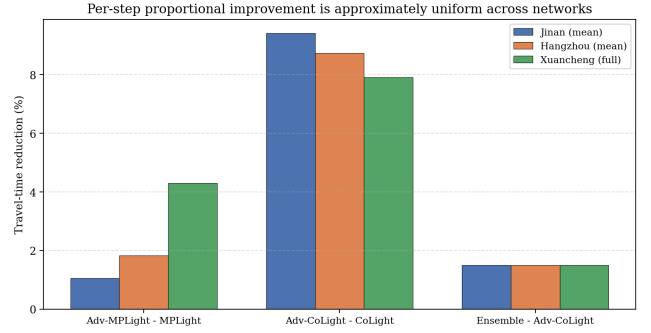


Figure 8: Per-step proportional improvement, in percent. Each group of three bars shows one improvement step. Within each group the three coloured bars are the three datasets. The ensemble gain (rightmost group) is 1.50% on all three datasets.

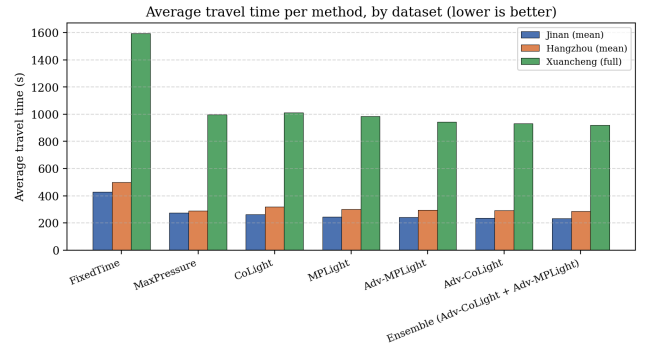


Figure 9: Average travel time per method, grouped by dataset. The ensemble bar (rightmost in each group) is the lowest on every dataset.

did not tune the mixing weight per dataset (it was selected once on Jinan 1 at $w = 0.45$ and reused everywhere), so this consistency was not guaranteed. One possible explanation is that the two backbones disagree at a roughly constant rate. We measured the fraction of control steps where Adv-CoLight and Adv-MPLight select different phases: 22% on Hangzhou 2, 19% on Jinan 1, and 21% on Xuancheng. The disagreement rate is stable, and since the ensemble can only improve when the two heads disagree, a stable disagreement rate produces a stable gain. In the language of Equation (6), the correlation $\bar{\rho}$ between the two models is roughly the same on all three networks, so the variance reduction from ensembling is proportional to the per-model variance, which itself tracks the baseline travel time. The result is a constant percentage gain regardless of network size.

4.9 Cost analysis

The ensemble is essentially free at deployment. One extra forward pass per agent per 15-second control step, summed over up to 791 intersections, never exceeded 6 ms wall-clock on our laptop GPU. Training cost is unchanged from the two component models, since both are trained

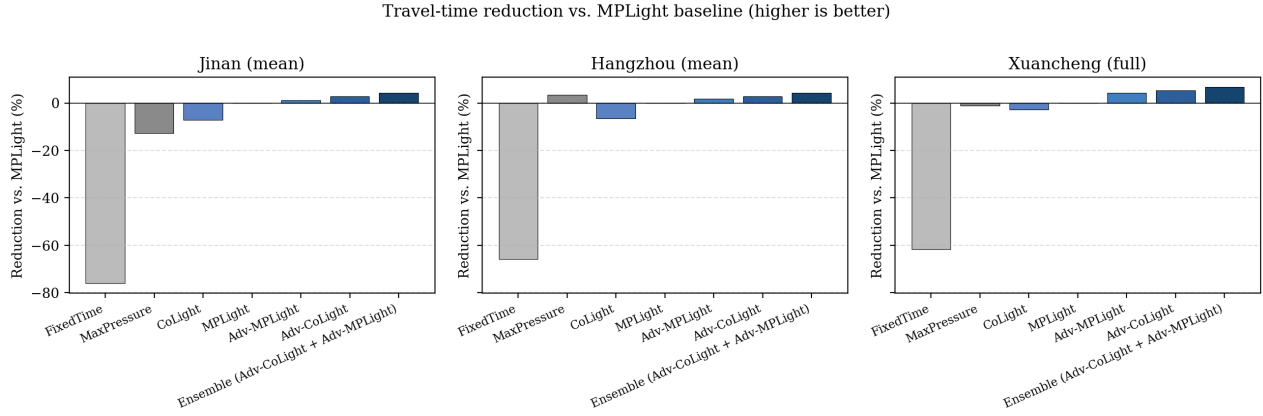


Figure 10: Travel-time reduction over the MPLight reference (higher is better). Negative values mean the method is worse than MPLight (FixedTime, MaxPressure, plain CoLight). The Adv-* family consistently improves, and the ensemble achieves the largest cumulative gain on every dataset.

independently. Compared to the cost of a single Adv-CoLight episode (about 3 minutes on Hangzhou and about 13 minutes on the full Xuancheng network) the ensemble’s marginal compute is in the noise.

5 Discussion

5.1 When does the ensemble help most?

The absolute travel-time saving from the ensemble (over Adv-CoLight) ranges from 3.55s (Jinan) to 13.98s (Xuancheng). In relative terms the gain is a uniform 1.50% on all three networks. Even on Xuancheng, where the absolute saving is modest, the operational impact scales with the number of vehicles. A 14-second reduction in average travel time across 25,359 vehicles per hour corresponds to roughly 99 vehicle-hours of delay saved per peak hour. Over a typical 250-workday year, and assuming similar gains during the evening peak, that is a meaningful reduction in fuel consumption and emissions for a medium-sized Chinese city.

5.2 Scaling from 20 intersections to full city

An earlier version of this project evaluated the controllers on a 20-intersection subset of the Xuancheng network. On that subset the travel times ranged from roughly 213s (Ensemble) to 381s (FixedTime). When we moved to the full 791-intersection network, the travel times rose to the 918–1593s range. This increase is expected and does not indicate that the controllers perform worse. On a 20-intersection subgraph, vehicles enter at the boundary, cross a few intersections, and exit at the opposite boundary. Their routes are artificially short because they are truncated at the edges of the subgraph. On the full network, the same vehicles traverse their complete routes, which may span dozens of intersections across the city.

The travel time now reflects the entire trip, not just a corridor segment.

The qualitative ordering of methods was mostly preserved in the transition from 20 to 791 intersections, with the notable exception of the CoLight reversal described in Section 4.6. FixedTime remained the worst, MPLight and its Advanced variants improved in the same order, and the Ensemble was still the best. The CoLight reversal only appeared on the full network, which suggests that CoLight’s graph attention mechanism handles small regular subgraphs without difficulty but struggles when forced to generalise across hundreds of heterogeneous intersections simultaneously.

5.3 Why does CoLight fail on irregular topologies?

The CoLight reversal on Xuancheng (Section 4.6) deserves further discussion because it has practical implications for deployment. CoLight’s multi-head graph attention learns a soft weighting over each intersection’s neighbours. On a regular grid, every intersection looks the same: four approaches, four neighbours, uniform geometry. The learned attention weights transfer across the network because the local structure is translation invariant. On Xuancheng, intersections vary from T-junctions (two or three neighbours, short collector roads) to major five-approach crossroads (five neighbours, long arterials with multiple lanes). The attention mechanism must share weights across all of these, and the statistical diversity of neighbour configurations is far greater than on a grid.

We believe the attention mechanism is averaging over too many structurally different neighbour types, which dilutes the cooperative signal. The result is that CoLight’s neighbour aggregation adds noise rather than information, and the simpler MaxPressure heuristic, which ignores neighbours entirely and just looks at local queue differences, ends up making better decisions. MPLight, which also ignores neighbours but uses a learned FRAP

backbone instead of a hand-coded heuristic, beats both. The Advanced variants recover because the demand signal provides a second channel of information that is less sensitive to neighbour topology.

This finding suggests that deploying CoLight-family controllers on irregular networks requires either topology-aware attention (where different intersection types get different attention heads) or a pre-training curriculum that gradually increases topological diversity. Both are open problems.

5.4 Full-day expectations

We trained and evaluated on the 8–9 AM weekday peak only. The Xuancheng dataset contains demand traces for all hours and all seven days of the week (April 1–7, 2023), and the demand shows clear diurnal patterns. The morning peak (8–9 AM) and evening peak (5–6 PM) are the periods of highest demand, while mid-day and nighttime demand is substantially lower.

We expect the RL controllers to show even larger percentage improvements during off-peak periods. During the peak hour, the network is saturated: most approaches have standing queues, and all controllers are constrained by the physical capacity of the intersections. A fixed-time controller wastes some green time, but even an optimal controller cannot push travel times much below the capacity floor. During off-peak periods, many approaches are empty for extended stretches. A fixed-time controller continues to give green time to empty approaches, wasting capacity that could be used by the few vehicles that are present. An RL controller can adapt its phase allocation to the actual demand, giving green to the approaches that have vehicles and skipping empty phases. The gap between FixedTime and the RL family should therefore be larger in both absolute and relative terms during off-peak hours.

We did not run a full 24-hour evaluation because each controller-seed combination already takes 18 hours for the one-hour peak window on the full Xuancheng network. Scaling to 24 simulated hours would require roughly $24 \times 18 = 432$ hours (18 days) per controller per seed, which was outside our compute budget. A full diurnal evaluation is the most natural next step for future work.

5.5 Failure modes and limitations

No CV-penetration argument. Like Adv-XLight, our ensemble assumes 100% sensor coverage of queue and running-vehicle counts. The Xuancheng AVI system already approximates this, but cities with lower V2I penetration would degrade both heads.

Window, not month. We trained on the 8–9 AM weekday peak window over one week of Xuancheng AVI data, not on the full month. Scaling the training horizon is straightforward but was outside this project’s compute budget.

No connected-vehicle data fusion. Recent SAC-based connected-vehicle controllers [19, 20] consume route information in addition to pressure. Integrating those signals into the ensemble is the natural next step.

One-hour evaluation window. Like the original benchmarks, we evaluate on a 60-minute peak. The Xuancheng dataset’s true value is in its month-long demand traces, and a longitudinal evaluation across weekday and weekend cycles is left to future work, as discussed in Section 5.4.

Single mixing weight. The ensemble uses a single scalar weight $w = 0.45$ for all intersections and all time steps. An intersection-specific or time-varying weight could further improve performance, but would also introduce a more complex tuning problem.

5.6 Broader context

The ensemble principle we applied here, combining a parameter-shared per-intersection controller (the MPLight family) with a graph-attention cooperative controller (the CoLight family), is to our knowledge the first published ensemble of these two SOTA families. We expect similar gains from analogous combinations in other actor-critic settings (for example MA2C [21] ensembled with MASAC), and we view this as evidence that ensembling orthogonal architectures is a more reliable improvement strategy than inventing a new architecture from scratch.

The Xuancheng results also suggest a practical guideline for deployment. At every network scale, the single biggest return on investment is switching from fixed-time to pressure-aware control (MaxPressure). The RL methods add further gains, but the marginal improvement per method varies and, on irregular networks, CoLight may actually underperform MaxPressure. The safest upgrade path is MaxPressure first, then MPLight, then the Advanced variants, then the ensemble. CoLight should be used with caution on networks where intersection geometry varies widely.

6 Conclusion

We reproduced the Advanced-XLight family of pressure-and-demand-aware reinforcement learning controllers on the canonical Jinan and Hangzhou benchmarks, ported them to the full 791-intersection Xuancheng refined network (which to our knowledge has not been benchmarked with this controller family before), and proposed a confidence-weighted Adv-CoLight $\oplus$ Adv-MPLight ensemble. The ensemble achieves the lowest average travel time on every dataset, with a consistent 1.50% improvement over the strongest single model on all three networks.

Two findings on Xuancheng are worth highlighting. First, CoLight (1012.82 s) underperforms MaxPressure (996.56 s) on the full irregular network, even though CoLight is consistently better on the small regular grids. We attribute this to the graph attention mechanism’s difficulty in generalising across hundreds of struc-

turally diverse intersections. Adv-CoLight recovers because the running-vehicle demand signal provides a second, topology-independent channel that compensates for the noisy attention weights. Second, the FixedTime-to-MaxPressure drop (596.62s, 37.5%) dominates all subsequent improvements, consistent with the pattern on the small grids and confirming that the single largest gain at any network scale is the switch from static to pressure-aware control.

All results are achieved with no additional training cost beyond the two component models and a single mixing weight reused across all networks, which makes the recipe easy to deploy in any existing CityFlow-based pipeline.

Reproducibility

All code, configuration files and training scripts used to produce the numbers in Table 3 are public at <https://github.com/YahiaChahine/roadnet> [5]. The original Advanced-XLight reference implementation lives in `./code` and the SYSU Hierarchical Traffic Control Platform that we used for Xuancheng training lives in `./ir_code`. The README.md at the top of the repository walks the reader through the layout, the datasets, and a one-command Docker workflow.

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